CS24311 — THEORY (3.0 CR) | MODULE V NOTES | BIT MESRA | NEP SCHEME

Module 5: Transformers & Generation Evaluation

Self-Attention, Multi-Head Attention, BERT vs GPT Architectures & BLEU / ROUGE Math

Module V: Transformers, Large Language Models & Generation Evaluation

1. Limitations of Recurrent Architectures (RNN, LSTM, GRU) & Sequential Bottleneck
2. The Transformer Architecture (Vaswani et al., 2017: "Attention Is All You Need")
3. Scaled Dot-Product Self-Attention Mechanism & Softmax Normalization ($\sqrt{d_k}$)
4. Multi-Head Attention Formulations ($\mathrm{Concat}(\mathrm{head}_1 \dots \mathrm{head}_h)W^O$)
5. Positional Encoding Schemes: Fixed Sinusoidal Waves vs. Rotary Embeddings (RoPE)
6. Transformer Encoder Blocks (LayerNorm, Residual Additions & Multi-Layer Feedforward)
7. Transformer Decoder Blocks (Masked Causal Self-Attention & Cross-Attention)
8. BERT Architecture: Bidirectional Transformer Encoders (Masked LM & NSP Tasks)
9. GPT Family: Autoregressive Decoder Models, In-Context Learning & Prompting
10. Text Generation Decoding: Greedy, Beam Search, Top-k & Top-p (Nucleus) Sampling
11. NLP Evaluation Metrics: BLEU Score (with Brevity Penalty) & ROUGE-1/2/L Math
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 2 – 4: The Transformer & Scaled Dot-Product Attention

1. Scaled Dot-Product Attention:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where $Q \in \mathbb{R}^{n \times d_k}$ is the Query matrix, $K \in \mathbb{R}^{m \times d_k}$ is the Key matrix, and $V \in \mathbb{R}^{m \times d_v}$ is the Value matrix.
Why Scale by $\frac{1}{\sqrt{d_k}}$? For large values of d_k, dot products grow large in magnitude, pushing the softmax function into regions with extremely small gradients. Dividing by $\sqrt{d_k}$ stabilizes gradient backpropagation!

2. Multi-Head Attention:

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \mathrm{head}_2, \dots, \mathrm{head}_h)W^O$$

$$\mathrm{head}_i = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

Where projection parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$, and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

Topic 8 & 9: BERT vs. GPT Architectural Comparison

Feature	BERT (Devlin et al., 2018)	GPT (Radford et al., OpenAI)
Core Architecture	Transformer **Encoder-Only** (Bidirectional attention).	Transformer **Decoder-Only** (Causal left-to-right masked attention).
Pre-Training Objectives	1. **Masked Language Model (MLM):** Predict 15% masked tokens. 2. **Next Sentence Prediction (NSP):** Binary classification.	**Autoregressive Language Modeling:** Predict next token $P(w_t \mid w_{
Context Flow	Full bidirectional visibility across future and past tokens.	Strictly unidirectional past context (future tokens masked out).
Primary Strength	NLU (Classification, Named Entity Recognition, Extractive QA).	NLG (Text Generation, Summarization, Code Synthesis, Chatbots).

Topic 11: Text Generation Evaluation Metrics (BLEU & ROUGE)

1. Bilingual Evaluation Understudy (BLEU) Metric:

$$\text{BLEU} = \text{BP} \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

Where p_n is the modified n-gram precision, and BP is the **Brevity Penalty** to penalize artificially short candidate outputs:

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r \end{cases}$$

Where c is candidate length and r is reference length.

🏛 Top BIT Mesra Exam Questions & Answers (Module V)

Q1. Explain why Self-Attention overcomes the fundamental sequential bottleneck of RNNs and LSTMs. (8 Marks)

1. **Massive Parallelization:** RNNs compute hidden state $h_t = f(h_{t-1}, x_t)$ sequentially, requiring $O(n)$ sequential time steps that cannot be parallelized across GPU cores. Self-Attention computes all pairwise token interactions simultaneously via matrix multiplications (QK^T) in $O(1)$ sequential operations.
2. **Direct Long-Range Paths:** In RNNs, information must travel step-by-step through n recurrent transitions, causing vanishing gradients for distant words. Self-Attention connects any two positions in the sequence with a direct $O(1)$ path length, enabling instantaneous long-range context resolution.